MU-HUA (MAURICE) WANG

mauricewang1128@gmail.com · +886 931 250 860 · **maurice1128.github.io** · Taipei, Taiwan

RESEARCH INTEREST

Physical assistive robotics for long-term care. I want to build robots that perform contact-rich care tasks on a bedridden body — starting with diaper changing, which demands three things at once: perceiving a body that will not cooperate, predicting how it moves when touched, and manipulating deformable material safely against skin. My work covers all three: a clinically validated markerless motion capture system, learned dynamics models of musculoskeletal bodies, and a measurement of whether compositional motion priors survive transfer to a robot hand.

EDUCATION

National Yang Ming Chiao Tung University — M.S., Biomedical Engineering

Entered Feb 2025 · on approved leave of absence since May 2025, enrollment retained · researching full time

National Yang Ming Chiao Tung University — B.S., Biomedical Engineering · Sep 2020 – Jun 2024

- GPA 3.72 / 4.30 · final two years 4.07 / 4.30 · **IELTS 8.0** (R 9.0, L 8.5, S 7.5, W 7.0)
- Deep Learning for Biomedical Data; Neural Engineering; Brain and Physiological Signal Analysis; Medical Device Regulatory Affairs; Smart Medical Device Mechanical Design

RESEARCH EXPERIENCE

Robot Learning and Embodied AI — independent · Oct 2025 – present

Sole author on all three; two submitted. Every result carries adversarial controls, multiple-comparison correction, and reports its own negative findings.

Two Routes to a Motor Prior: Guidance versus a Frozen Body Model — *submitted, TMLR*

- A frozen babble-learned body model against a distilled teacher policy under one SAC backbone on MyoSuite bodies, with **seven controls** and a matched withdrawal at a common training step.
- Guidance buys the early speed (−50.6 mrad elbow, −11.7 mm finger) but its lead reverses inside the budget on one body. n = 12 matched seeds, Holm–Bonferroni across six stated family definitions.

Which Hand-Motion Composition Axes Are Worth Measuring — *submitted, IEEE RA-L*

- Built the measurement that modular motion priors assume but rarely test: a paired split holding out compositions, six axes over three hand datasets, five gates, plus a raw-data interaction statistic computable before any model is trained.
- **Finding: compositional difficulty appears only where the two factors interact** — every axis carrying it has positive interaction, none carries it without. Scoped as one-sided evidence, since only one of the six axes could have falsified the direction.
- Reported against myself: the statistic detects presence well but magnitude badly, and on a Shadow Hand 16 of 17 comparisons return null.

What Master-Then-Reorganise Does Not Buy: An Invariance Argument for Body Codes

- With the core frozen and only a body code θ adapted, $\theta \to M\theta$ with $W\theta M^{-1}$ preserves the function, so adaptation under an equivariant rule cannot change held-out error.
- Built a re-encoding that is the identity at every training code — preserving leave-one-body-out decodability by construction (0.000 across 20 seeds) — yet still costs +0.875 mm: a counterexample to decodability as a proxy for representation quality.

RGB Multi-View 3D Markerless Motion Capture — NYCU · Oct 2023 – Sep 2025

Research Assistant, then Team Lead · Advisor: Prof. Hui-Ting Shih

Real-time whole-body pose sensing for clinical use — the perception layer a robot working over a bed needs. I wrote the pipeline, ran the validation, and collected the clinical data.

- **Built the real-time pipeline end to end**: GPU acceleration and CPU parallelism across capture, inference, triangulation and inverse kinematics; DLT with occlusion handling; and **ROWV**, an outlier-rejection algorithm beating KNN and RANSAC on this problem.

- **Validated it**: VICON comparison over 21 participants, plus 104 walking trials from 11 participants on a nine-camera ring.
- **Asked how much sensing error the downstream decision can absorb** — forward-dynamic simulation with a Geyer–Herr reflex controller over 66 design cells: identification of a weak muscle group falls 0.845 → 0.549 as joint-angle error grows 0° → 12°, and mild weakness is never identified reliably at any error level.
- **Ran the collaborations and the field work**: 2 hospital MOUs (Taipei Veterans General; Taipei City Hospital), 200+ stroke and knee-pain patients in Vietnam, 40+ at Kaohsiung Chang Gung, 20+ elderly at a daycare centre, amputee trials with NYCU Physical Therapy. Wards and clinics, not a lab.
- **Shipped the product**, not only the algorithm: PyQt6 clinician GUI, SQL and cloud pipeline with ICD-10 coding, Unreal Engine rehabilitation game. Led a 5-person team; 6 media features.

Behavioural Trajectory Modelling — Taoyuan General Hospital · Jan 2023 – present

Advisor: Dr. Tso-Jeng Wang

Mapped methadone dose trajectories to a polar angle so shape is comparable independent of magnitude, then clustered with a bivariate von Mises mixture on the torus. 9,900 urine tests from 552 patients; reported what did not survive.

EEG and EMG Sleep–Epilepsy Analysis — NYCU · Oct 2022 – Oct 2023

Advisor: Prof. You-Yin Chen

Built a pipeline processing 24-hour recordings in 3 minutes, with automatic WAKE / NREM / REM staging from band-specific energy features and a manual-scoring GUI for cross-labeler consistency. Designed a surgical modification that raised EMG signal-to-noise at the source.

PUBLICATIONS

Peer-reviewed conference papers

1. **Wang, M.-H.**, et al. *Markerless Gait Analysis for Lower Back Pain: Assessing Pelvis–Trunk Coordination.* SEMBA 2025. **(first author)**
2. Huang, L.-W., **Wang, M.-H.**, et al. *A Real-Time Algorithm for Detecting Gait Events in Lower-Limb Prosthesis Users.* ASB 2025.
3. Huang, A.-Y., **Wang, M.-H.**, et al. *Spatiotemporal Cortical-Synergy Connectivity in the Gait Cycle.* SEMBA 2025.
4. Chang, C.-W., **Wang, M.-H.**, et al. *Central Thalamic Deep Brain Stimulation and Sleep in an Alzheimer's Model.* ISMRM 2024.
5. Chen, B.-H., et al. incl. **Wang, M.-H.** *Robust Self-Occlusion Mitigation in Multi-View 3D Markerless Motion Capture.* ICMST 2024.

Submitted. Wang, M.-H. (sole author). *Which Hand-Motion Composition Axes Are Worth Measuring.* **IEEE Robotics and Automation Letters.** — **Wang, M.-H.** (sole author). *Two Routes to a Motor Prior: Guidance Versus a Frozen Body Model.* **Transactions on Machine Learning Research.**

Manuscript drafts (available on request). One further sole-author robot-learning paper; four in human motion sensing and clinical biomechanics; one in behavioural data science.

TEACHING

Physical Science, Grade 8 — Tamkang High School · Feb – Aug 2026

Class 8 rose to 2nd of eight classes, its margin over the school average widening +5.1 → +11.8 points across three term exams. **The gain came from the bottom**: its bottom-12% cutoff finished 28 points above the school's, up from 14, with the second-narrowest spread in the grade.

SKILLS AND AWARDS

Languages / ML — Python, MATLAB, C/C++ · PyTorch · SAC, Dyna model-based RL, policy distillation, world models · MyoSuite, Shadow Hand, inverse kinematics, DLT triangulation

Systems / Statistics — CuPy, Numba, GPU acceleration, PyQt6, SQL, ICD-10 · matched-seed paired-*t*, Holm–Bonferroni, von Mises directional statistics, adversarial controls, pre-registration

Awards — 2nd Place, Youth AI Startup Hackathon, Ministry of Digital Affairs (2023) · 1st Place, Elderly Health & Technology, NCNU (2023)